

ISLA Gaia

Computer Engineering

Operating Systems Architecture

Final Project

A Linux Enterprise Environment

Wanderful Travels, Lda.

wanderful.local • 192.168.56.0/24 • Ubuntu Server 22.04 LTS

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1. Executive Summary

Wanderful Travels, Lda. is a fictional travel agency based in Braga, Portugal, with 10 employees. The company specialises in organising customised travel packages across Europe and requires a reliable, secure IT infrastructure to support daily operations, internal file sharing, and a public-facing website.

This project implements that infrastructure across three virtual machines running Ubuntu Server 22.04 LTS, hosted in Oracle VirtualBox on a Windows host. VM1 (srv-infra) provides internal network services including DNS resolution, DHCP address assignment, and Samba file sharing. VM2 (srv-web) hosts the company website as a WordPress installation served over HTTPS through Apache. VM3 (srv-client) acts as a client workstation and remote backup target, receiving its IP address automatically from the DHCP server on srv-infra.

Key architectural decisions include the use of BIND9 for authoritative DNS over the wonderful.local domain, isc-dhcp-server for address assignment, Samba for file sharing accessible from both Linux clients and the Windows host, MariaDB as the database engine for WordPress, and OpenVPN for encrypted remote access from the Windows host into the internal subnet. Backups are automated via a bash script

2. Architecture Diagram

All three machines share the Host-Only network (vboxnet0) on subnet 192.168.56.0/24. Each machine also has a NAT adapter for outbound internet access during installation.

Component	Details
Subnet	192.168.56.0/24 (Host-Only vboxnet0)
srv-infra	192.168.56.10 — DNS, DHCP, Samba, OpenVPN
srv-web	192.168.56.20 — Apache, MariaDB, PHP, WordPress
srv-client	192.168.56.50–100 (DHCP) — client, backup target
Windows host	192.168.56.1 — browser, VPN client, Samba client

3. Machine Identification

All three virtual machines were created from scratch for this project. The table below summarises the identification data collected immediately after the first boot of each machine, before any configuration was applied.

Field	srv-infra	srv-web	srv-client
Hostname	srv-infra	srv-web	srv-client
Role	Infra server	Web server	Client / backup
IP address	192.168.56.10	192.168.56.20	DHCP
OS	Ubuntu 22.04 LTS	Ubuntu 22.04 LTS	Ubuntu 22.04 LTS
Kernel (uname -r)	5.15.0-181-generic	5.15.0-119-generic	5.15.0-181-generic
MAC (enp0s8)	08:00:27:73:71:9b	08:00:27:af:88:1f	08:00:27:62:03:fa
VirtualBox UUID	823c2380-f612-4cf2-a6c6-d62f3f77a49e	fb9a7e8f-1506-417a-9800-ee48c1a51de0	a869a081-4ab8-473e-b7d9-593f64c4bef4

List of commands ran to gather information:

```
uname -r  
ip link show enp0s8 | grep ether  
cat /etc/os-release
```

cat /etc/os-release output — srv-infra

```
sonia@srv-infra:~$ cat /etc/os-release  
PRETTY_NAME="Ubuntu 22.04.5 LTS"  
NAME="Ubuntu"  
VERSION_ID="22.04"  
VERSION="22.04.5 LTS (Jammy Jellyfish)"  
VERSION_CODENAME=jammy  
ID=ubuntu  
ID_LIKE=debian  
HOME_URL="https://www.ubuntu.com/"  
SUPPORT_URL="https://help.ubuntu.com/"  
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"  
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"  
UBUNTU_CODENAME=jammy
```

cat /etc/os-release output — srv-web

```
sonia@srv-web:~$ cat /etc/os-release
PRETTY_NAME="Ubuntu 22.04.5 LTS"
NAME="Ubuntu"
VERSION_ID="22.04"
VERSION="22.04.5 LTS (Jammy Jellyfish)"
VERSION_CODENAME=jammy
ID=ubuntu
ID_LIKE=debian
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"
UBUNTU_CODENAME=jammy
```

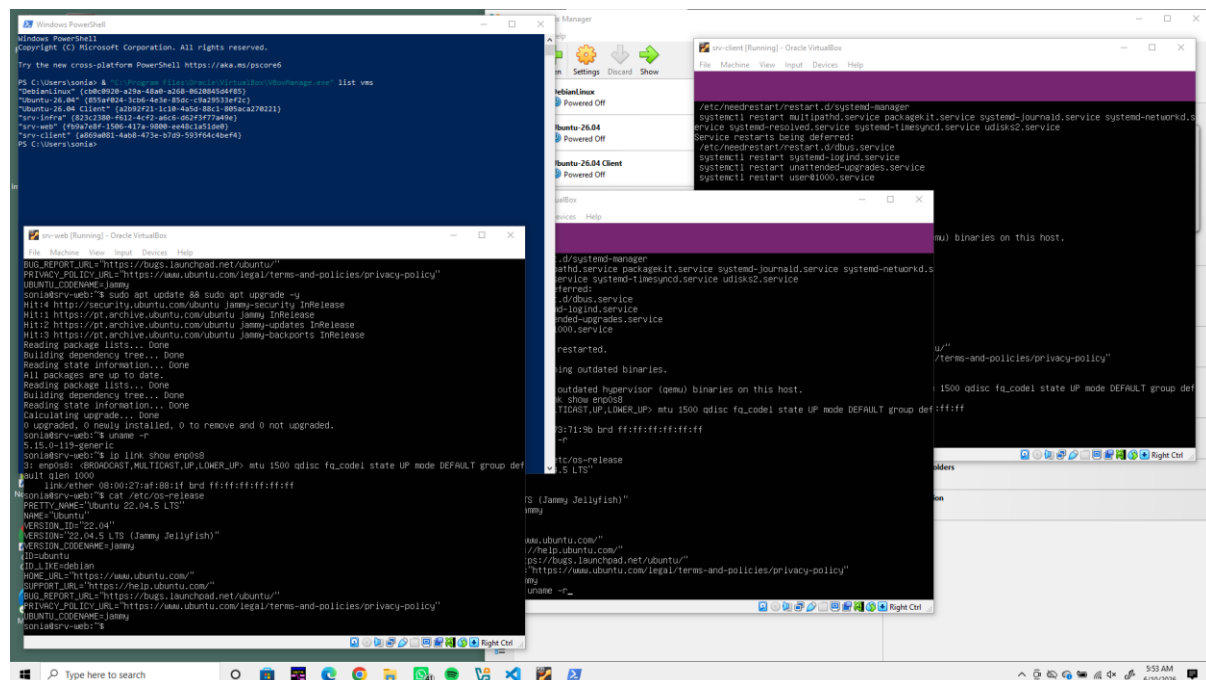
cat /etc/os-release output — srv-client

```
sonia@srv-client:~$ cat /etc/os-release
PRETTY_NAME="Ubuntu 22.04.5 LTS"
NAME="Ubuntu"
VERSION_ID="22.04"
VERSION="22.04.5 LTS (Jammy Jellyfish)"
VERSION_CODENAME=jammy
ID=ubuntu
ID_LIKE=debian
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"
UBUNTU_CODENAME=jammy
```

After gathering information regarding the 3 VMs (after the first boot), i updated all VMs using this command:

```
sudo apt update && sudo apt upgrade -y
```

VirtualBox dashboard screenshot



4. Service Configuration

4.1. Networking and Firewall

VM: srv-infra and srv-web (static IPs) — srv-client (DHCP)

Goal

Assign static IP addresses to srv-infra and srv-web so that services are always reachable at a known address. Configure UFW on all three machines with a default-deny inbound policy and explicit rules for each open service. Populate /etc/hosts on all VMs for name resolution before DNS is configured.

Decisions

Netplan was used to configure static IPs as it is the standard network configuration tool on Ubuntu 22.04. UFW was chosen for its simplicity and readable rule syntax. Static IPs were assigned to infrastructure and web servers; the client was left on DHCP to demonstrate that the DHCP server works.

Commands executed — srv-infra

```
sudo nano /etc/netplan/01-netcfg.yaml
sudo netplan apply
sudo hostnamectl set-hostname srv-infra
sonia@srv-infra:~$ ip addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:c1:0e:dc brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 metric 100 brd 10.0.2.255 scope global dynamic enp0s3
        valid_lft 86355sec preferred_lft 86355sec
    inet6 fd17:625c:f037:2:a00:27ff:fec1:edc/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 86356sec preferred_lft 14356sec
    inet6 fe80::a00:27ff:fec1:edc/64 scope link
        valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:73:71:9b brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.10/24 brd 192.168.56.255 scope global enp0s8
        valid_lft forever preferred_lft forever
    inet 192.168.56.105/24 metric 100 brd 192.168.56.255 scope global secondary dynamic enp0s8
        valid_lft 555sec preferred_lft 555sec
    inet6 fe80::a00:27ff:fe73:719b/64 scope link
        valid_lft forever preferred_lft forever
sudo ufw default deny incoming
sudo ufw default allow outgoing
sudo ufw allow 22/tcp
sudo ufw enable
```

Netplan configuration — srv-infra (/etc/netplan/01-netcfg.yaml)

 srv-infra [Running] - Oracle VirtualBox

File Machine View Input Devices Help

```
GNU nano 6.2 /etc/netplan/01-netcfg.yaml *
network:
  version: 2
  ethernets:
    enp0s8:
      dhcp4: no
      addresses: [192.168.56.10/24]
```

Commands executed — srv-web

```
sudo nano /etc/netplan/01-netcfg.yaml # use 192.168.56.20
sudo netplan apply
sudo hostnamectl set-hostname srv-web
sonia@srv-web:~$ ip addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:a7:f9:fe brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 metric 100 brd 10.0.2.255 scope global dynamic enp0s3
        valid_lft 86392sec preferred_lft 86392sec
    inet6 fd17:625c:f037:2:a00:27ff:fea7:f9fe/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 86393sec preferred_lft 14393sec
    inet6 fe80::a00:27ff:fea7:f9fe/64 scope link
        valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:af:88:1f brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.20/24 brd 192.168.56.255 scope global enp0s8
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:feaf:881f/64 scope link
        valid_lft forever preferred_lft forever
sonia@srv-web:~$
sudo ufw default deny incoming
sudo ufw default allow outgoing
sudo ufw allow 22/tcp
sudo ufw allow 80/tcp
sudo ufw allow 443/tcp
sudo ufw enable
```

/etc/hosts — applied to all 3 VMs

```
192.168.56.10    srv-infra wonderful.local
192.168.56.20    srv-web
```

 srv-client [Running] - Oracle VirtualBox

```
File Machine View Input Devices Help
GNU nano 6.2 /etc/hosts *
127.0.0.1 localhost
127.0.1.1 srv-client

# The following lines are desirable for IPv6 capable hosts
::1          ip6-localhost ip6-loopback
fe00::0      ip6-localnet
ff00::0      ip6-mcastprefix
ff02::1      ip6-allnodes
ff02::2      ip6-allrouters

# host-only network
192.168.56.10 srv-infra wonderful.local
192.168.56.20 srv-web
```

Testing the host configuration

```
sonia@srv-web:~$ ping -c2 wonderful.local
PING srv-infra (192.168.56.10) 56(84) bytes of data.
64 bytes from srv-infra (192.168.56.10): icmp_seq=1 ttl=64 time=1.51 ms
64 bytes from srv-infra (192.168.56.10): icmp_seq=2 ttl=64 time=1.92 ms

--- srv-infra ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1001ms
rtt min/avg/max/mdev = 1.510/1.713/1.916/0.203 ms
sonia@srv-web:~$

sonia@srv-web:~$ ping -c2 srv-web
PING srv-web (127.0.1.1) 56(84) bytes of data.
64 bytes from srv-web (127.0.1.1): icmp_seq=1 ttl=64 time=0.082 ms
64 bytes from srv-web (127.0.1.1): icmp_seq=2 ttl=64 time=0.037 ms

--- srv-web ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1011ms
rtt min/avg/max/mdev = 0.037/0.059/0.082/0.022 ms
sonia@srv-infra:~$ ping -c2 srv-web
ping: srv-web: Temporary failure in name resolution
sonia@srv-infra:~$ ping -c2 wonderful.local
PING srv-infra (192.168.56.10) 56(84) bytes of data.
64 bytes from srv-infra (192.168.56.10): icmp_seq=1 ttl=64 time=0.060 ms
64 bytes from srv-infra (192.168.56.10): icmp_seq=2 ttl=64 time=0.039 ms

--- srv-infra ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1022ms
rtt min/avg/max/mdev = 0.039/0.049/0.060/0.010 ms

sonia@srv-infra:~$ ping -c2 srv-web
PING srv-web (192.168.56.20) 56(84) bytes of data.
64 bytes from srv-web (192.168.56.20): icmp_seq=1 ttl=64 time=0.854 ms
64 bytes from srv-web (192.168.56.20): icmp_seq=2 ttl=64 time=0.953 ms

--- srv-web ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1001ms
rtt min/avg/max/mdev = 0.854/0.903/0.953/0.049 ms
```

Screenshot — UFW status on each VM

```
sonia@srv-infra:~$ sudo ufw status verbose
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), disabled (routed)
New profiles: skip
```

To	Action	From
--	-----	----
22/tcp	ALLOW IN	Anywhere
22/tcp (v6)	ALLOW IN	Anywhere (v6)

```
sonia@srv-web:~$ sudo ufw status verbose
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), disabled (routed)
New profiles: skip
```

To	Action	From
--	-----	----
22/tcp	ALLOW IN	Anywhere
80/tcp	ALLOW IN	Anywhere
443/tcp	ALLOW IN	Anywhere
22/tcp (v6)	ALLOW IN	Anywhere (v6)
80/tcp (v6)	ALLOW IN	Anywhere (v6)
443/tcp (v6)	ALLOW IN	Anywhere (v6)

```
sonia@srv-client:~$ sudo ufw status verbose
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), disabled (routed)
New profiles: skip
```

To	Action	From
--	-----	----
22/tcp	ALLOW IN	Anywhere
22/tcp (v6)	ALLOW IN	Anywhere (v6)

4.2. Users and SSH

VM: all three

Goal

Create at least 3 non-root users on each VM, one with sudo privileges. Configure key-based SSH authentication using ed25519 keys between all machines and from the Windows host. Disable password authentication on srv-web to demonstrate hardened SSH configuration.

Decisions

ed25519 was chosen over RSA as it produces shorter keys with equivalent or better security. ssh-copy-id was used to distribute public keys. Password authentication was disabled only on srv-web as required, after confirming that key-based login worked.

Commands executed — on each VM

```
sudo useradd -m -s /bin/bash ana
sudo passwd ana
sudo useradd -m -s /bin/bash mihai
sudo passwd mihai
sudo useradd -m -s /bin/bash corina
sudo passwd corina
sudo usermod -aG sudo ana # grant sudo to ana
```

Generate key on Windows host (PowerShell)

```
ssh-keygen -t ed25519 -C "windows-host"
type C:\Users\sonia\.ssh\id_ed25519.pub | ssh ana@192.168.56.10 "mkdir -p ~/.ssh &&
cat >> ~/.ssh/authorized_keys"
type C:\Users\sonia\.ssh\id_ed25519.pub | ssh ana@192.168.56.20 "mkdir -p ~/.ssh &&
cat >> ~/.ssh/authorized_keys"
type C:\Users\sonia\.ssh\id_ed25519.pub | ssh ana@192.168.56.108 "mkdir -p ~/.ssh
&& cat >> ~/.ssh/authorized_keys" # after srv-client gets DHCP IP
```

Key-based SSH between VMs — on srv-infra

```
ssh-keygen -t ed25519
ssh-copy-id ana@192.168.56.20
ssh-copy-id ana@192.168.56.108
```

Disable password auth on srv-web (/etc/ssh/sshd_config)

```
PasswordAuthentication no
PubkeyAuthentication yes
```

```
sudo systemctl restart ssh
```

Screenshot — successful SSH login

```

PS C:\Users\sonia> type C:\Users\sonia\.ssh\id_ed25519.pub | ssh ana@192.168.56.10 "mkdir -p ~/.ssh && cat >> ~/.ssh/authorized_keys"
The authenticity of host '192.168.56.10 (192.168.56.10)' can't be established.
ED25519 key fingerprint is SHA256:hJNKfQ2wYNiZ6sTGukFvlyWbTQTnnr8B1woU+989K08.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.56.10' (ED25519) to the list of known hosts.
ana@192.168.56.10's password:
PS C:\Users\sonia> ssh ana@192.168.56.10
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 5.15.0-181-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Wed Jun 10 06:46:56 AM UTC 2026

System load:            0.0
Usage of /:             21.7% of 24.44GB
Memory usage:          14%
Swap usage:            0%
Processes:             105
Users logged in:       1
IPv4 address for enp0s3: 10.0.2.15
IPv6 address for enp0s3: fd17:625c:f037:2:a00:27ff:fec1:edc

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.

   https://ubuntu.com/engage/secure-kubernetes-at-the-edge

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

New release '24.04.4 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ana@srv-infra:~$ exit
logout
Connection to 192.168.56.10 closed.
PS C:\Users\sonia> type C:\Users\sonia\.ssh\id_ed25519.pub | ssh ana@192.168.56.20 "mkdir -p ~/.ssh && cat >> ~/.ssh/authorized_keys"
The authenticity of host '192.168.56.20 (192.168.56.20)' can't be established.
ED25519 key fingerprint is SHA256:a+ZHC+XdjCjF1AVrgyqPdOkC1ldcyVcTajuKf/Ta/8.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.56.20' (ED25519) to the list of known hosts.
ana@192.168.56.20's password:
PS C:\Users\sonia> ssh ana@192.168.56.20
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 5.15.0-119-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Wed Jun 10 06:51:14 AM UTC 2026

System load:            0.0
Usage of /:             24.8% of 24.44GB
Memory usage:          15%
Swap usage:            0%
Processes:             107
Users logged in:       1
IPv4 address for enp0s3: 10.0.2.15
IPv6 address for enp0s3: fd17:625c:f037:2:a00:27ff:fea7:f9fe

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.

   https://ubuntu.com/engage/secure-kubernetes-at-the-edge

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

New release '24.04.4 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

*** System restart required ***
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ana@srv-web:~$
ana@srv-web:~$ exit
logout
Connection to 192.168.56.20 closed.

```

```
PS C:\Users\sonia> ssh ana@192.168.56.108
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 5.15.0-181-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Wed Jun 10 10:43:36 AM UTC 2026

System load:          0.01
Usage of /:            21.7% of 24.44GB
Memory usage:         14%
Swap usage:           0%
Processes:            108
Users logged in:      1
IPv4 address for enp0s3: 10.0.2.15
IPv6 address for enp0s3: fd17:625c:f037:2:a00:27ff:fedb:b3b7

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.

   https://ubuntu.com/engage/secure-kubernetes-at-the-edge

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

New release '24.04.4 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ana@srv-client: $
```

4.3. DNS (BIND9) and DHCP

VM: srv-infra

Goal

Run an authoritative DNS server for the wonderful.local domain with a forward zone and a reverse zone for 192.168.56.0/24. Run a DHCP server that delivers an IP address, default gateway, and DNS server address to clients joining the network. Demonstrate that srv-client receives an address via DHCP and that name resolution works.

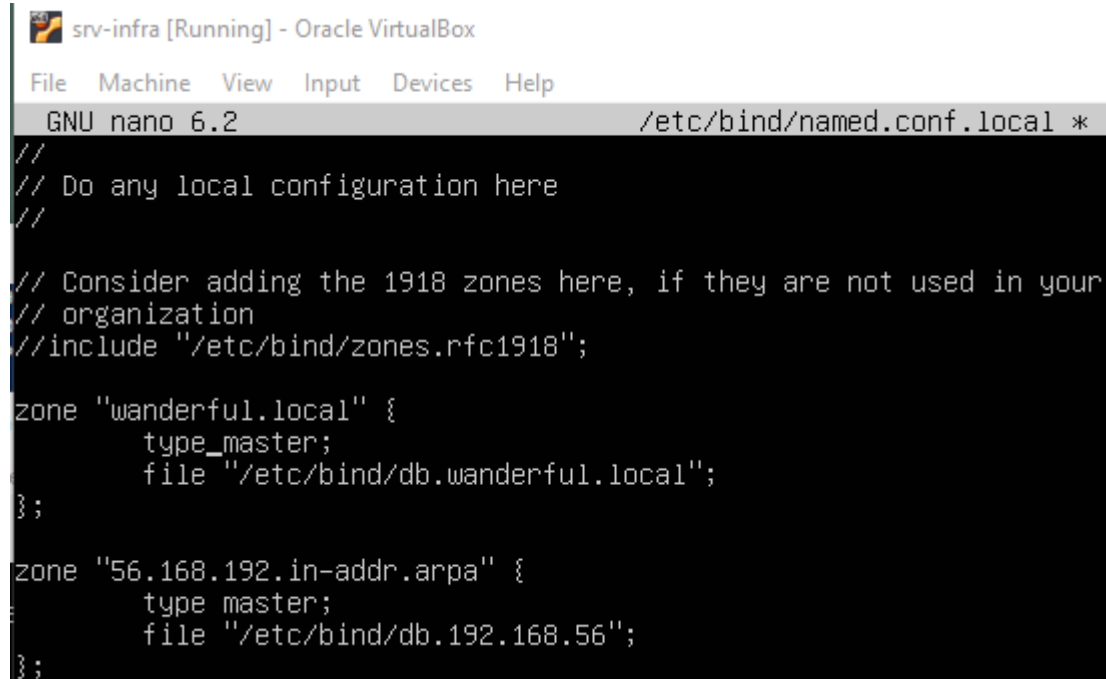
Decisions

BIND9 was chosen as it is the standard authoritative DNS implementation on Ubuntu and was covered in class. isc-dhcp-server was chosen for DHCP as it integrates naturally with BIND9 and is widely documented.

Installation

```
sudo apt install -y bind9 bind9utils isc-dhcp-server
```

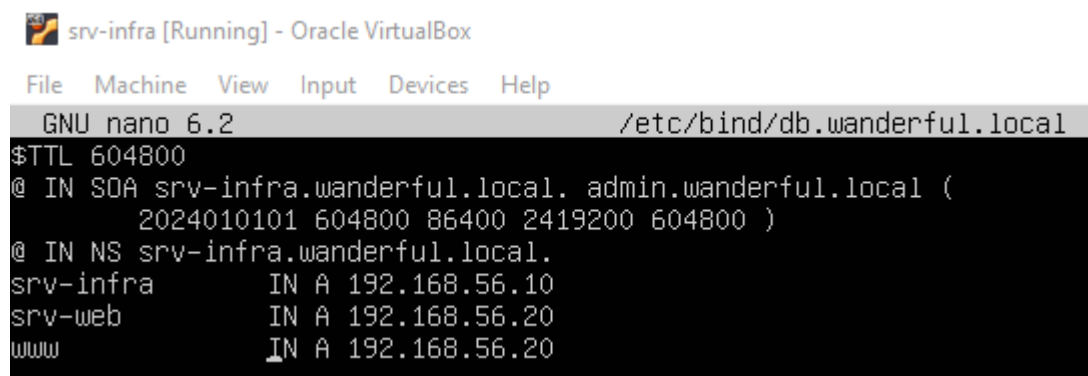
/etc/bind/named.conf.local



The screenshot shows a terminal window titled "srv-infra [Running] - Oracle VirtualBox". The terminal displays the GNU nano 6.2 editor editing /etc/bind/named.conf.local. The configuration includes comments for local configuration and RFC1918 zones, and two zone definitions: "wonderful.local" as a master zone pointing to /etc/bind/db.wonderful.local, and "56.168.192.in-addr.arpa" as a master zone pointing to /etc/bind/db.192.168.56.

```
//  
// Do any local configuration here  
//  
  
// Consider adding the 1918 zones here, if they are not used in your  
// organization  
//include "/etc/bind/zones.rfc1918";  
  
zone "wonderful.local" {  
    type_master;  
    file "/etc/bind/db.wonderful.local";  
};  
  
zone "56.168.192.in-addr.arpa" {  
    type master;  
    file "/etc/bind/db.192.168.56";  
};
```

/etc/bind/db.wonderful.local (forward zone)



The screenshot shows a terminal window titled "srv-infra [Running] - Oracle VirtualBox". The terminal displays the GNU nano 6.2 editor editing /etc/bind/db.wonderful.local. The configuration sets a TTL of 604800 and defines SOA, NS, and A records for the wonderful.local domain.

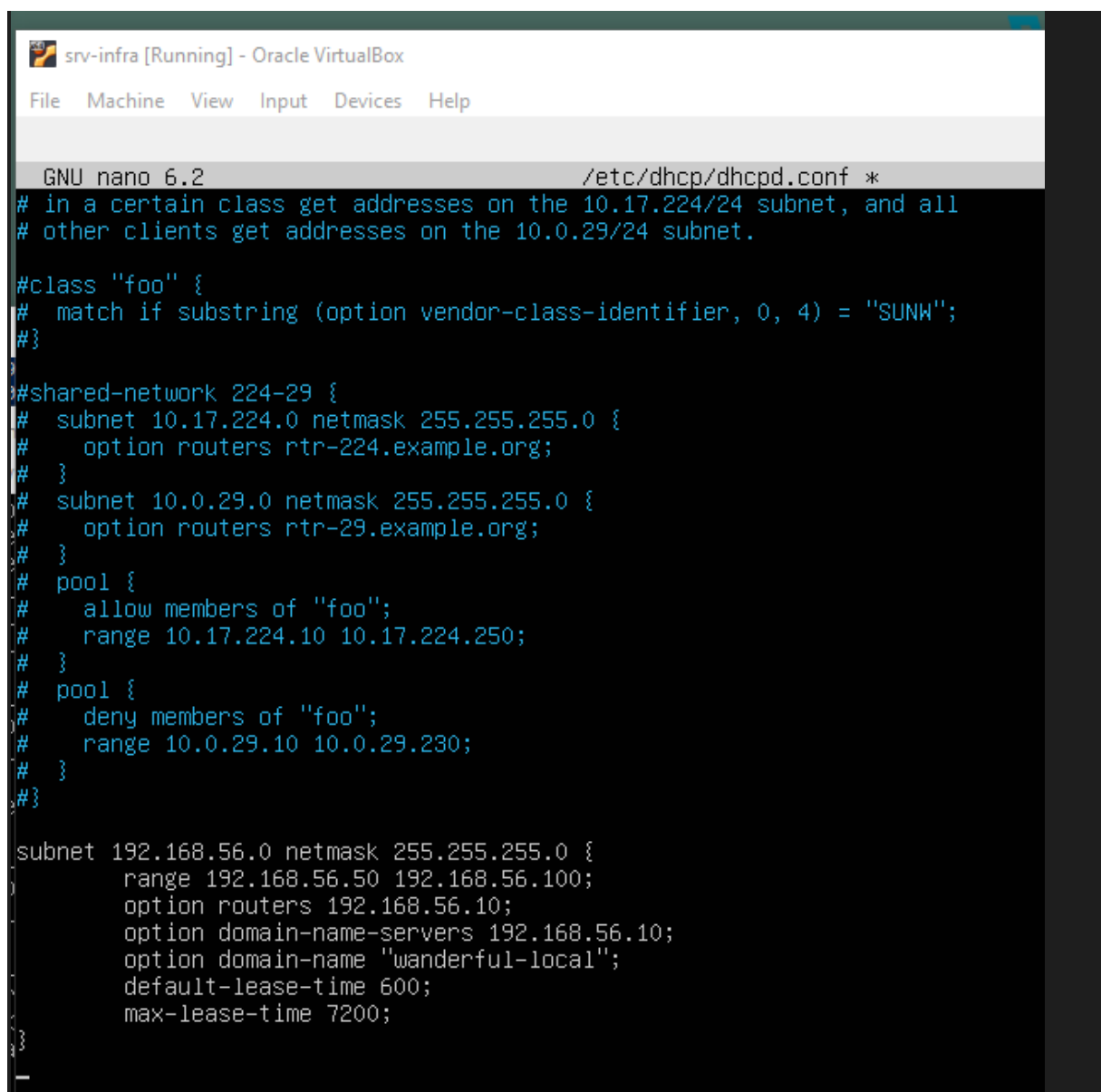
```
$TTL 604800  
@ IN SOA srv-infra.wonderful.local. admin.wonderful.local (  
    2024010101 604800 86400 2419200 604800 )  
@ IN NS srv-infra.wonderful.local.  
srv-infra      IN A 192.168.56.10  
srv-web        IN A 192.168.56.20  
www            IN A 192.168.56.20
```

/etc/bind/db.192.168.56 (reverse zone) srv-infra [Running] - Oracle VirtualBox

File Machine View Input Devices Help

```
GNU nano 6.2 /etc/bind/db.192.168.56
$TTL 604800
@ IN SOA srv-infra.wanderful.local. admin.wanderful.local. (
    2024010101 604800 86400 2419200 604800 )
@ IN NS srv-infra.wanderful.local.
10 IN PTR srv-infra.wanderful.local.
20 IN PTR srv-web.wanderful.local.
```

/etc/dhcp/dhcpd.conf



```
GNU nano 6.2 /etc/dhcp/dhcpd.conf *
# in a certain class get addresses on the 10.17.224/24 subnet, and all
# other clients get addresses on the 10.0.29/24 subnet.

#class "foo" {
#  match if substring (option vendor-class-identifier, 0, 4) = "SUNW";
#}

#shared-network 224-29 {
#  subnet 10.17.224.0 netmask 255.255.255.0 {
#    option routers rtr-224.example.org;
#  }
#  subnet 10.0.29.0 netmask 255.255.255.0 {
#    option routers rtr-29.example.org;
#  }
#  pool {
#    allow members of "foo";
#    range 10.17.224.10 10.17.224.250;
#  }
#  pool {
#    deny members of "foo";
#    range 10.0.29.10 10.0.29.230;
#  }
#}

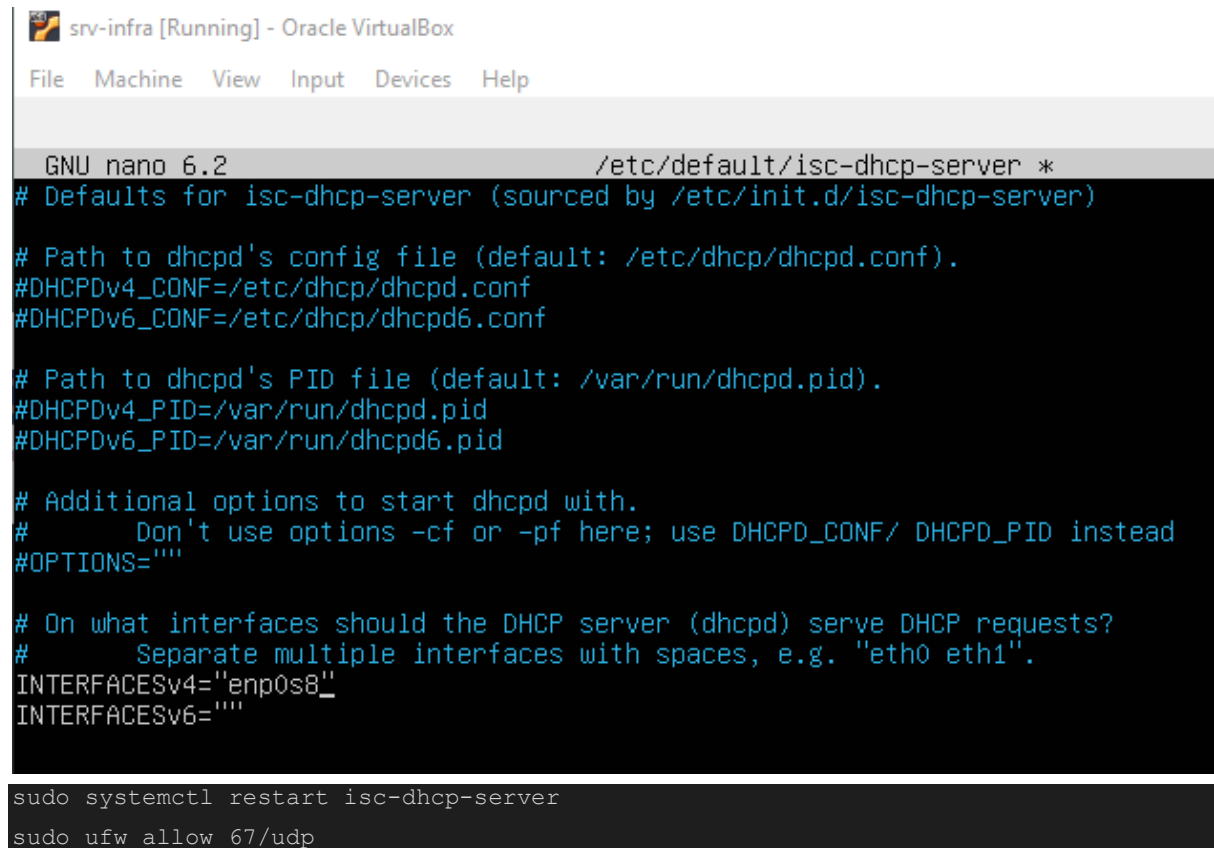
subnet 192.168.56.0 netmask 255.255.255.0 {
  range 192.168.56.50 192.168.56.100;
  option routers 192.168.56.10;
  option domain-name-servers 192.168.56.10;
  option domain-name "wanderful-local";
  default-lease-time 600;
  max-lease-time 7200;
}
```

Start services and open firewall

```
sudo named-checkconf
sudo named-checkzone wanderful.local /etc/bind/db.wanderful.local
sudo systemctl enable --now bind9
sudo systemctl enable --now isc-dhcp-server
sudo ufw allow 53
sudo ufw allow 67/udp
```

Set DHCP interface and start

```
sudo nano /etc/default/isc-dhcp-server
```



```
srv-infra [Running] - Oracle VirtualBox
File Machine View Input Devices Help

GNU nano 6.2 /etc/default/isc-dhcp-server *
# Defaults for isc-dhcp-server (sourced by /etc/init.d/isc-dhcp-server)

# Path to dhcpd's config file (default: /etc/dhcp/dhcpd.conf).
#DHCPDv4_CONF=/etc/dhcp/dhcpd.conf
#DHCPDv6_CONF=/etc/dhcp/dhcpd6.conf

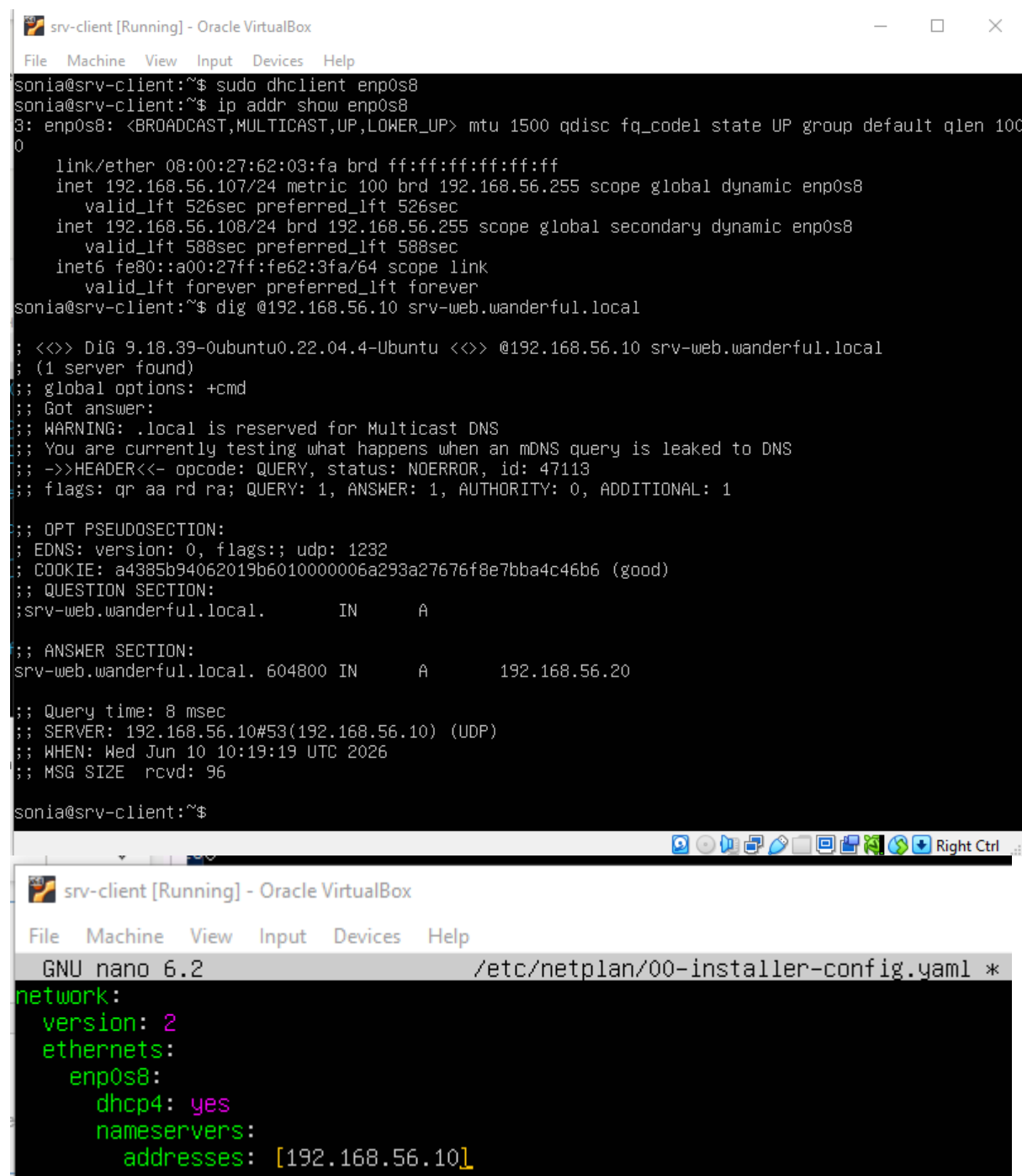
# Path to dhcpd's PID file (default: /var/run/dhcpd.pid).
#DHCPDv4_PID=/var/run/dhcpd.pid
#DHCPDv6_PID=/var/run/dhcpd6.pid

# Additional options to start dhcpd with.
# Don't use options -cf or -pf here; use DHCPD_CONF/ DHCPD_PID instead
#OPTIONS=""

# On what interfaces should the DHCP server (dhcpd) serve DHCP requests?
# Separate multiple interfaces with spaces, e.g. "eth0 eth1".
INTERFACESv4="enp0s8"
INTERFACESv6=""

sudo systemctl restart isc-dhcp-server
sudo ufw allow 67/udp
```

Screenshot - Test DHCP on srv-client



```

srv-client [Running] - Oracle VirtualBox
File Machine View Input Devices Help
sonia@srv-client:~$ sudo dhclient enp0s8
sonia@srv-client:~$ ip addr show enp0s8
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:62:03:fa brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.107/24 metric 100 brd 192.168.56.255 scope global dynamic enp0s8
        valid_lft 526sec preferred_lft 526sec
    inet 192.168.56.108/24 brd 192.168.56.255 scope global secondary dynamic enp0s8
        valid_lft 588sec preferred_lft 588sec
    inet6 fe80::a00:27ff:fe62:3fa/64 scope link
        valid_lft forever preferred_lft forever
sonia@srv-client:~$ dig @192.168.56.10 srv-web.wanderful.local

; <<>> DiG 9.18.39-0ubuntu0.22.04.4-Ubuntu <<>> @192.168.56.10 srv-web.wanderful.local
; (1 server found)
;; global options: +cmd
;; Got answer:
;; WARNING: .local is reserved for Multicast DNS
;; You are currently testing what happens when an mDNS query is leaked to DNS
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 47113
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: a4385b94062019b6010000006a293a27676f8e7bba4c46b6 (good)
;; QUESTION SECTION:
;srv-web.wanderful.local.      IN      A

;; ANSWER SECTION:
srv-web.wanderful.local. 604800 IN      A      192.168.56.20

;; Query time: 8 msec
;; SERVER: 192.168.56.10#53(192.168.56.10) (UDP)
;; WHEN: Wed Jun 10 10:19:19 UTC 2026
;; MSG SIZE rcvd: 96

sonia@srv-client:~$

```



```

srv-client [Running] - Oracle VirtualBox
File Machine View Input Devices Help
GNU nano 6.2 /etc/netplan/00-installer-config.yaml *
network:
  version: 2
  ethernets:
    enp0s8:
      dhcp4: yes
      nameservers:
        addresses: [192.168.56.10]

```

```
sonia@srv-client:~$ ping -c 2 srv-web.waderful.local
ping: srv-web.waderful.local: Temporary failure in name resolution
sonia@srv-client:~$ ping -c 2 srv-web.wanderful.local
ping: srv-web.wanderful.local: Temporary failure in name resolution
sonia@srv-client:~$ sudo chmod 600 /etc/netplan/00-installer-config.yaml
sonia@srv-client:~$ sudo netplan apply
WARNING:root:Cannot call Open vSwitch: ovsdb-server.service is not running.
sonia@srv-client:~$ resolvectl status
Global
  Protocols: -LLMNR -mDNS -DNSoverTLS DNSSEC=no/unsupported
  resolv.conf mode: stub

Link 2 (enp0s3)
  Current Scopes: DNS
  Protocols: +DefaultRoute +LLMNR -mDNS -DNSoverTLS DNSSEC=no/unsupported
  DNS Servers: 10.239.63.27

Link 3 (enp0s8)
  Current Scopes: DNS
  Protocols: +DefaultRoute +LLMNR -mDNS -DNSoverTLS DNSSEC=no/unsupported
  Current DNS Server: 192.168.56.10
  DNS Servers: 192.168.56.10
sonia@srv-client:~$ sudo rm /etc/resolv.conf
sonia@srv-client:~$ sudo ln -s
ln: missing file operand
Try 'ln --help' for more information.
sonia@srv-client:~$ sudo ln -s /run/systemd/resolve/resolv.conf /etc/resolv.conf
sonia@srv-client:~$ sudo systemctl restart systemd-resolved
sonia@srv-client:~$ ping -c 2 srv-web.wanderful.local
PING srv-web.wanderful.local (192.168.56.20) 56(84) bytes of data.
64 bytes from srv-web (192.168.56.20): icmp_seq=1 ttl=64 time=4.75 ms
64 bytes from srv-web (192.168.56.20): icmp_seq=2 ttl=64 time=2.17 ms

--- srv-web.wanderful.local ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1002ms
rtt min/avg/max/mdev = 2.173/3.462/4.752/1.289 ms
sonia@srv-client:~$
```

Screenshot — DNS resolution with dig

```

srv-client [Running] - Oracle VirtualBox
File Machine View Input Devices Help
password: password updated successfully
sonia@srv-client:~$ sudo usermod -aG sudo ana
sonia@srv-client:~$ dig @192.168.56.10 srv-web.wanderland.local
;; communications error to 192.168.56.10#53: timed out
;; communications error to 192.168.56.10#53: timed out
;; communications error to 192.168.56.10#53: timed out

<<>> DiG 9.18.39-0ubuntu0.22.04.4-Ubuntu <<>> @192.168.56.10 srv-web.wander
(1 server found)
;; global options: +cmd
;; no servers could be reached
sonia@srv-client:~$ dig @192.168.56.10 srv-web.wanderful.local

<<>> DiG 9.18.39-0ubuntu0.22.04.4-Ubuntu <<>> @192.168.56.10 srv-web.wander
(1 server found)
;; global options: +cmd
;; Got answer:
;; WARNING: .local is reserved for Multicast DNS
;; You are currently testing what happens when an mDNS query is leaked to DNS
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 35031
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: 478a25135c73ecfb010000006a291e7c292113c1ddc1f917 (good)
;; QUESTION SECTION:
srv-web.wanderful.local.      IN      A

;; ANSWER SECTION:
srv-web.wanderful.local. 604800 IN      A      192.168.56.20

;; Query time: 0 msec
;; SERVER: 192.168.56.10#53(192.168.56.10) (UDP)
;; WHEN: Wed Jun 10 08:21:16 UTC 2026
;; MSG SIZE rcvd: 96

sonia@srv-client:~$

```

Screenshot - dig @192.168.56.10 srv-web.wanderful.local (run from srv-client after DHCP)

```

sonia@srv-client:~$ ip addr show enp0s8
8: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:62:03:fa brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.107/24 metric 100 brd 192.168.56.255 scope global dynamic enp0s8
        valid_lft 548sec preferred_lft 548sec
    inet6 fe80::a00:27ff:fe62:3fa/64 scope link
        valid_lft forever preferred_lft forever
sonia@srv-client:~$ _

```

4.4. Samba File Sharing

VM: srv-infra

Goal

Create one public Samba share accessible without credentials, and one authenticated share accessible only by the user ana. Demonstrate access from srv-client (Linux) and from the Windows host.

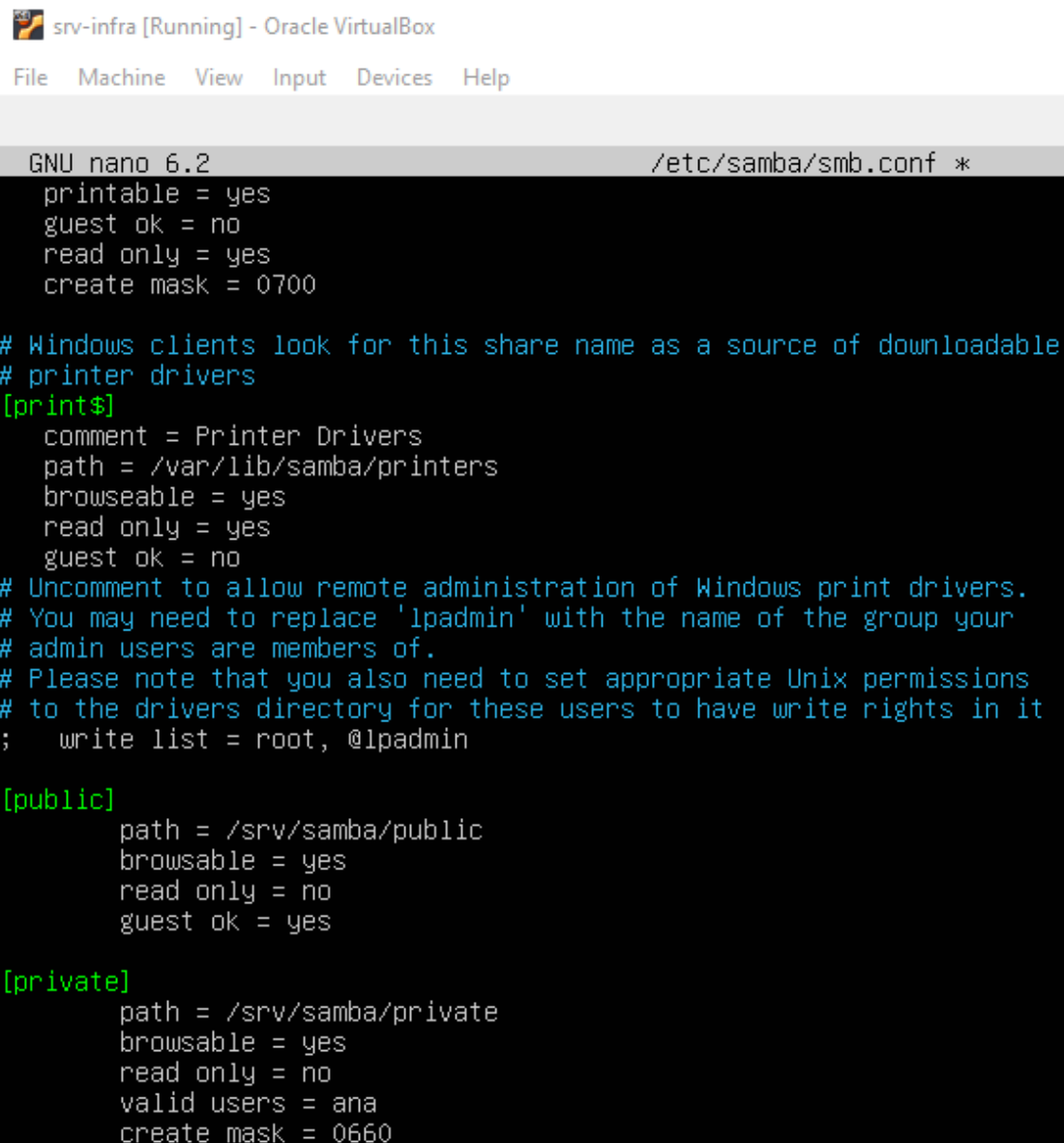
Decisions

Samba was chosen as it is the standard interoperability layer between Linux and Windows file sharing. Two shares were created to demonstrate both guest and authenticated access modes, as required by the project specification.

Installation and directory setup

```
sudo apt install -y samba
sudo mkdir -p /srv/samba/public
sudo mkdir -p /srv/samba/private
sudo chmod 0777 /srv/samba/public
sudo chmod 0770 /srv/samba/private
sudo chgrp sambashare /srv/samba/private
```

/etc/samba/smb.conf (relevant sections)



```
GNU nano 6.2 /etc/samba/smb.conf *
printable = yes
guest ok = no
read only = yes
create mask = 0700

# Windows clients look for this share name as a source of downloadable
# printer drivers
[print$]
comment = Printer Drivers
path = /var/lib/samba/printers
browseable = yes
read only = yes
guest ok = no
# Uncomment to allow remote administration of Windows print drivers.
# You may need to replace 'lpadmin' with the name of the group your
# admin users are members of.
# Please note that you also need to set appropriate Unix permissions
# to the drivers directory for these users to have write rights in it
; write list = root, @lpadmin

[public]
path = /srv/samba/public
browsable = yes
read only = no
guest ok = yes

[private]
path = /srv/samba/private
browsable = yes
read only = no
valid users = ana
create mask = 0660
```

Set Samba password and start service

```
sudo smbpasswd -a ana #anasmb
sudo smbpasswd -e ana
sudo systemctl enable --now smbd nmbd
sudo ufw allow samba
```

Test from srv-client

```
sudo apt install -y smbclient
smbclient //192.168.56.10/public -N
smbclient //192.168.56.10/private -U ana
```

Screenshot — Samba access

```
sonia@srv-client:~$ smbclient //192.168.56.10/public -N
Try "help" to get a list of possible commands.
smb: \> ls
.                D          0   Wed Jun 10 11:52:49 2026
..               D          0   Wed Jun 10 11:53:02 2026

                25623780 blocks of size 1024. 18604332 blocks available
smb: \>

sonia@srv-client:~$ smbclient //192.168.56.10/private -U ana
Password for [WORKGROUP\ana]:
Try "help" to get a list of possible commands.
smb: \> ls
.                D          0   Wed Jun 10 11:53:02 2026
..               D          0   Wed Jun 10 11:53:02 2026

                25623780 blocks of size 1024. 18604328 blocks available
smb: \> _
```

4.5. LAMP Stack and WordPress

VM: srv-web

Goal

Install Apache, MariaDB, and PHP (the LAMP stack) on srv-web. Create a dedicated database and a dedicated database user with no global privileges. Install WordPress, select a theme, and create an About page and two posts relevant to Wanderful Travels, Lda.

Decisions

MariaDB was chosen over MySQL as it is the default on Ubuntu 22.04 and is fully compatible with WordPress. The WordPress database user was granted privileges only to the wordpress_db database.

Installation

```
sudo apt install -y apache2 mariadb-server php php-mysql php-curl php-gd php-mbstring php-xml php-zip libapache2-mod-php
```

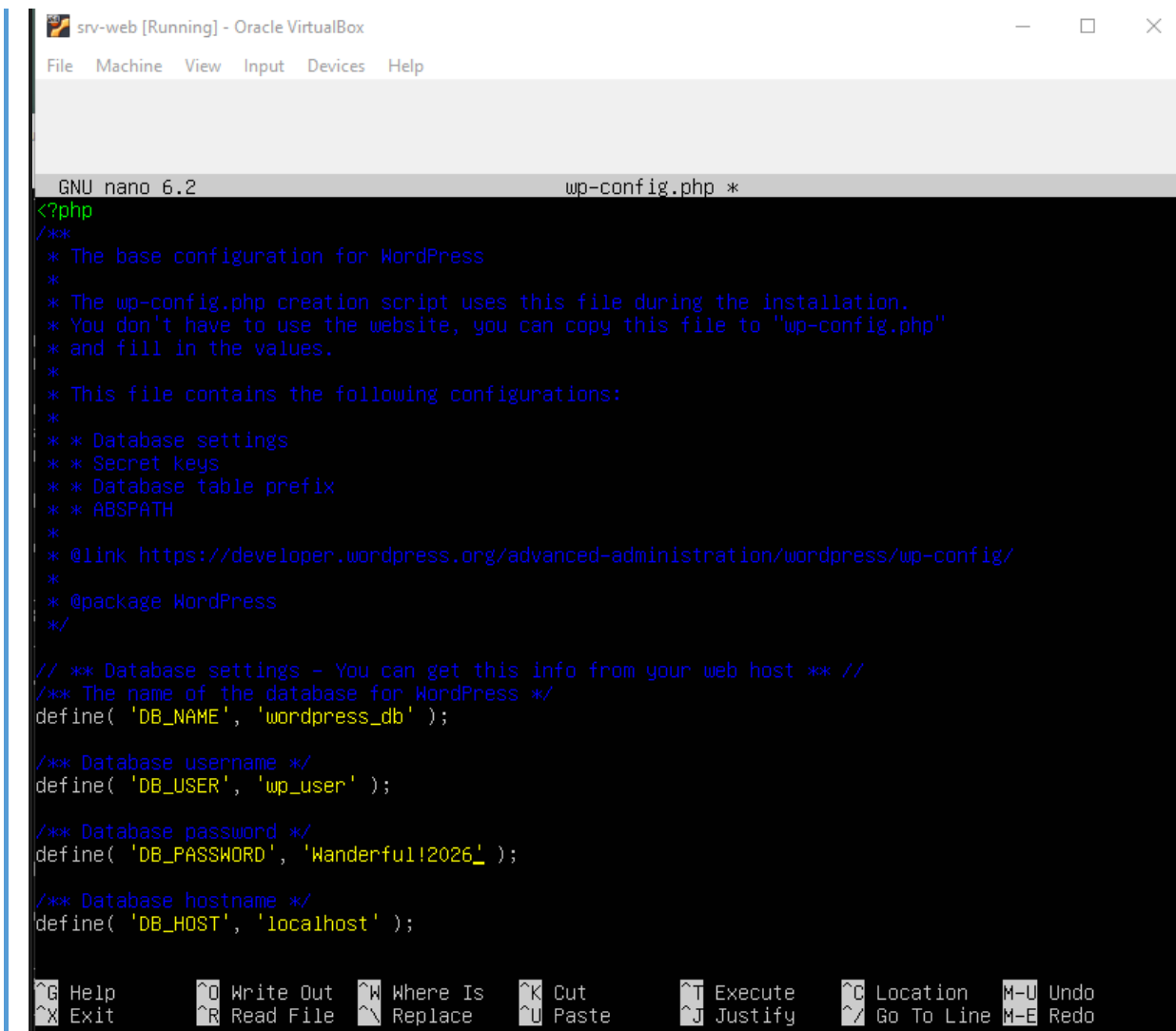
Secure MariaDB and create database

```
sudo mysql_secure_installation

sudo mysql -u root -p
CREATE DATABASE wordpress_db;
CREATE USER 'wp_user'@'localhost' IDENTIFIED BY 'Wanderful!2026';
GRANT ALL PRIVILEGES ON wordpress_db.* TO 'wp_user'@'localhost';
FLUSH PRIVILEGES;
EXIT;
```

Install WordPress

```
cd /tmp
wget https://wordpress.org/latest.tar.gz
tar xzf latest.tar.gz
sudo mv wordpress /var/www/wanderful.local
sudo chown -R www-data:www-data /var/www/wanderful.local
cd /var/www/wanderful.local
sudo cp wp-config-sample.php wp-config.php
sudo nano wp-config.php
```



The screenshot shows a window titled 'srv-web [Running] - Oracle VirtualBox'. Inside, the GNU nano 6.2 editor is open, editing the file 'wp-config.php *'. The editor contains PHP code for WordPress configuration. Comments explain the file's purpose and list configurations: Database settings, Secret keys, Database table prefix, and ABSPATH. It also includes a link to the WordPress documentation and package information. The configuration section defines database settings: DB_NAME (wordpress_db), DB_USER (wp_user), DB_PASSWORD (Wanderful!2026_), and DB_HOST (localhost). The bottom of the window shows a toolbar with various editing shortcuts.

```
GNU nano 6.2 wp-config.php *
<?php
/**
 * The base configuration for WordPress
 *
 * The wp-config.php creation script uses this file during the installation.
 * You don't have to use the website, you can copy this file to "wp-config.php"
 * and fill in the values.
 *
 * This file contains the following configurations:
 *
 * * Database settings
 * * Secret keys
 * * Database table prefix
 * * ABSPATH
 *
 * @link https://developer.wordpress.org/advanced-administration/wordpress/wp-config/
 *
 * @package WordPress
 */

// ** Database settings - You can get this info from your web host ** //
/** The name of the database for WordPress */
define( 'DB_NAME', 'wordpress_db' );

/** Database username */
define( 'DB_USER', 'wp_user' );

/** Database password */
define( 'DB_PASSWORD', 'Wanderful!2026_' );

/** Database hostname */
define( 'DB_HOST', 'localhost' );

^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^_ Go To Line  M-E Redo
```

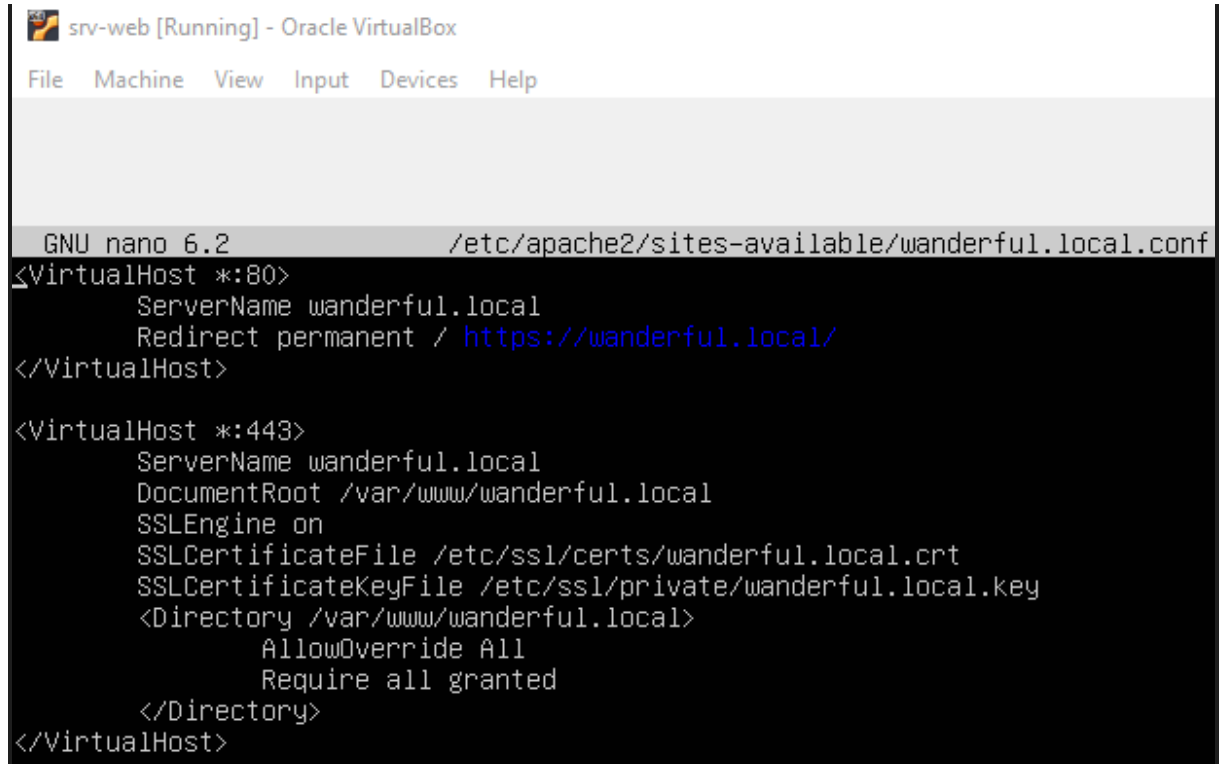
```
sonia@srv-web:/var/www/wanderful.local$ sudo a2enmod rewrite
[sudo] password for sonia:
Enabling module rewrite.
To activate the new configuration, you need to run:
  systemctl restart apache2
sonia@srv-web:/var/www/wanderful.local$ sudo a2ensite wanderful.local.conf
ERROR: Site wanderful.local does not exist!
sonia@srv-web:/var/www/wanderful.local$ sudo a2dissite 000-default.conf
Site 000-default disabled.
To activate the new configuration, you need to run:
  systemctl reload apache2
sonia@srv-web:/var/www/wanderful.local$ sudo systemctl restart apache2
sonia@srv-web:/var/www/wanderful.local$
```

Screenshot — WordPress installed

The screenshot shows the WordPress installation success screen. At the top is the WordPress logo. Below it, a large box contains the text "Success!" followed by "WordPress has been installed. Thank you, and enjoy!". It then displays the "Username" as "admin" and the "Password" as "Your chosen password.". A "Log In" link is provided at the bottom of this box.

Below the success message is the WordPress dashboard. The dashboard has a dark sidebar on the left with a menu including Home, Updates, Posts, Media, Pages, Comments, Appearance, Plugins, Users, Tools, Settings, and a Collapse Menu option. The main content area is titled "Dashboard" and features a "Welcome to WordPress!" banner with a link to "Learn more about the 7.0 version.". Below the banner are three cards: "Author rich content with blocks and patterns", "Customize your entire site with block themes", and "Switch up your site's look & feel with Styles".

At the bottom of the dashboard, there are several widgets. The "Site Health Status" widget shows "No information yet...". The "Quick Draft" widget has fields for "Title" and "Content" (with the placeholder "What's on your mind?") and a "Save Draft" button. The "At a Glance" widget shows "1 Published post", "1 Published page", and "1 Comment". The "WordPress Events and News" widget is also visible at the bottom.



```
GNU nano 6.2 /etc/apache2/sites-available/wonderful.local.conf
<VirtualHost *:80>
    ServerName wonderful.local
    Redirect permanent / https://wonderful.local/
</VirtualHost>

<VirtualHost *:443>
    ServerName wonderful.local
    DocumentRoot /var/www/wonderful.local
    SSLEngine on
    SSLCertificateFile /etc/ssl/certs/wonderful.local.crt
    SSLCertificateKeyFile /etc/ssl/private/wonderful.local.key
    <Directory /var/www/wonderful.local>
        AllowOverride All
        Require all granted
    </Directory>
</VirtualHost>
```

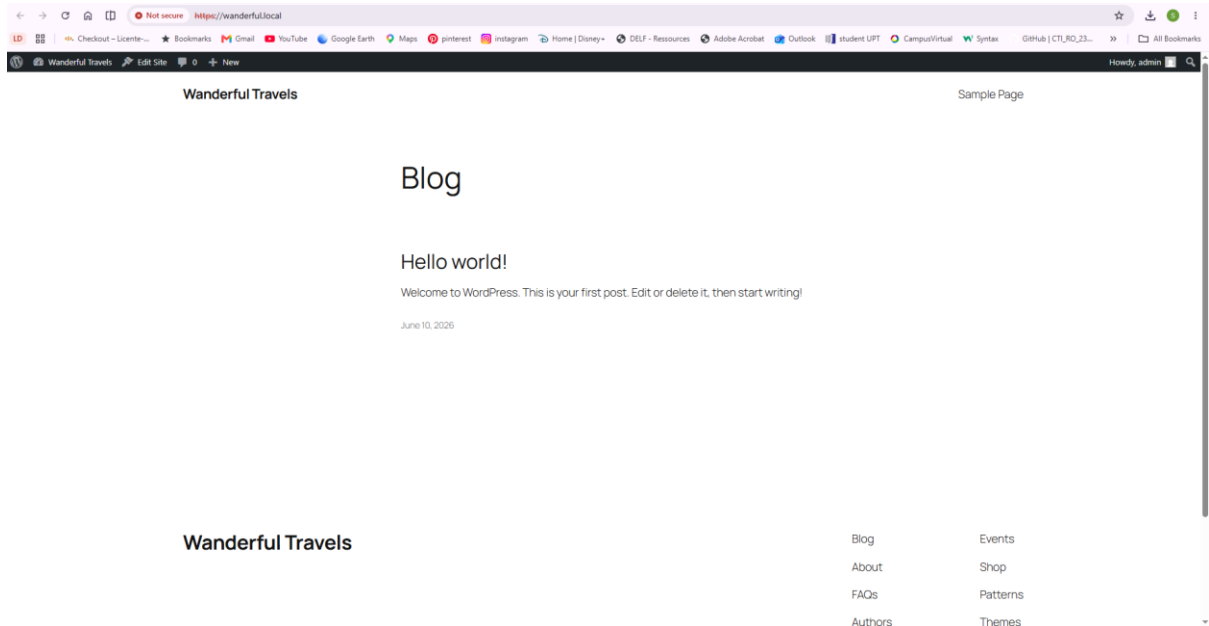
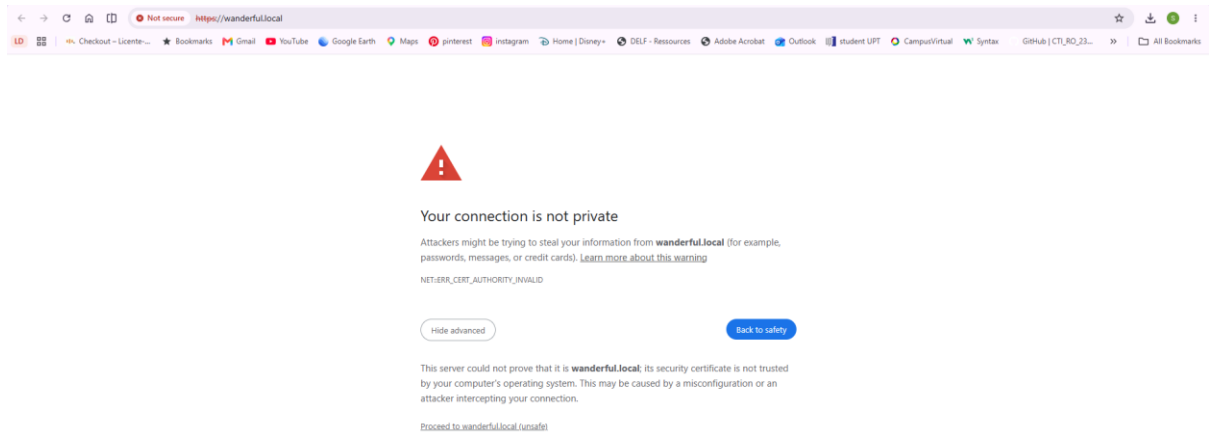
Enable modules and site

```
sudo a2enmod ssl rewrite
sudo a2ensite wonderful.local.conf
sudo a2dissite 000-default.conf
sudo systemctl restart apache2
```

Add to Windows hosts file (C:\Windows\System32\drivers\etc\hosts)

```
192.168.56.20 wonderful.local
```

Screenshot — HTTPS working



```
PS C:\Users\sonia> curl.exe -I http://wanderful.local
HTTP/1.1 301 Moved Permanently
Date: Thu, 11 Jun 2026 16:54:34 GMT
Server: Apache/2.4.52 (Ubuntu)
Location: https://wanderful.local/
Content-Type: text/html; charset=iso-8859-1
```

4.7. OpenVPN

VM: srv-infra

Goal

Install and configure OpenVPN on srv-infra to allow the Windows host to connect to the 192.168.56.0/24 internal subnet through an encrypted tunnel. Once connected, the Windows host should be able to access WordPress on srv-web using the internal IP address, without modifying the Windows hosts file.

Decisions

OpenVPN was chosen as it is the most widely supported open-source VPN solution and is covered in the course. easy-rsa was used to manage the PKI (Certificate Authority, server certificate, and client certificates) as it is the standard companion tool for OpenVPN.

Installation and PKI setup

```
sudo apt install -y openvpn easy-rsa
make-cadir ~/openvpn-ca
cd ~/openvpn-ca
./easyrsa init-pki
./easyrsa build-ca nopass
./easyrsa gen-req server nopass
./easyrsa sign-req server server
./easyrsa gen-dh
openvpn --genkey secret ta.key
./easyrsa gen-req client1 nopass
./easyrsa sign-req client client1
```

/etc/openvpn/server.conf



srv-infra [Running] - Oracle VirtualBox

File Machine View Input Devices Help

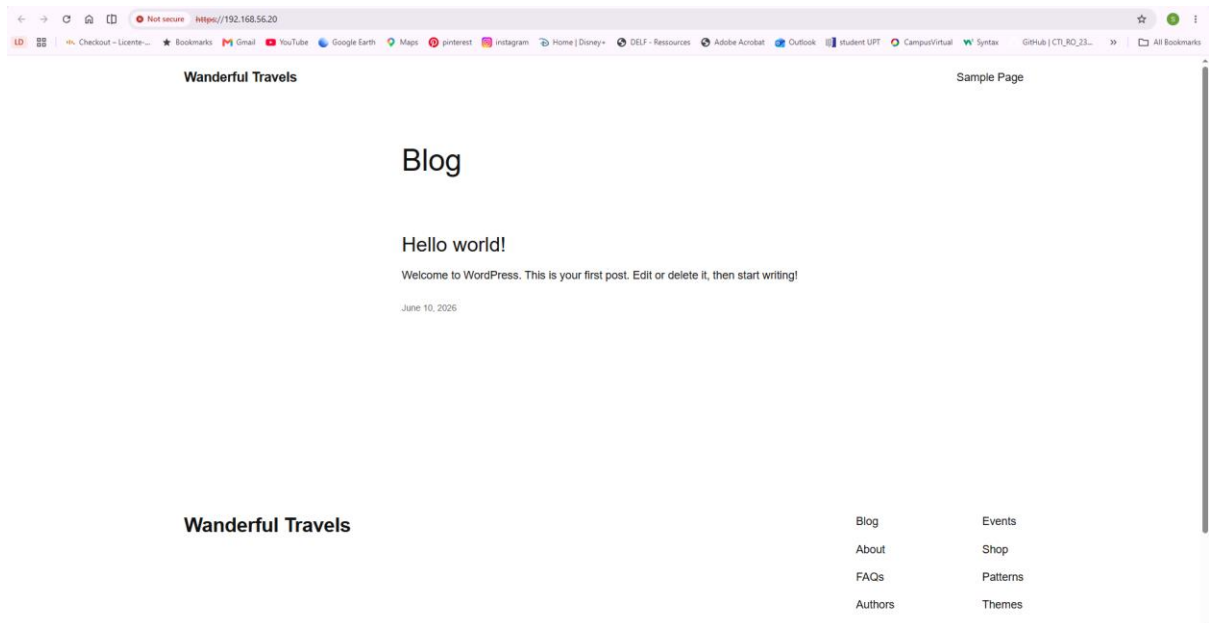
```
GNU nano 6.2 /etc/openvpn/server.conf *
port 1194
proto udp
dev tun
ca /root/openvpn-ca/pki/ca.crt
cert /root/openvpn-ca/pki/issued/server.crt
key /root/openvpn-ca/pki/private/server.key
dh /root/openvpn-ca/pki/dh.pem
tls-auth /root/openvpn-ca/ta.key 0
server 10.8.0.0 255.255.255.0
push "route 192.168.56.0 255.255.255.0"
keepalive 10 120
comp-lzo
persist-key
persist-tun
```

Start OpenVPN and open firewall

```
sudo systemctl enable --now openvpn@server
```

```
sudo ufw allow 1194/udp
```

Screenshot — VPN connected



4.8. Backup, Automation, and Restore

VM: srv-web

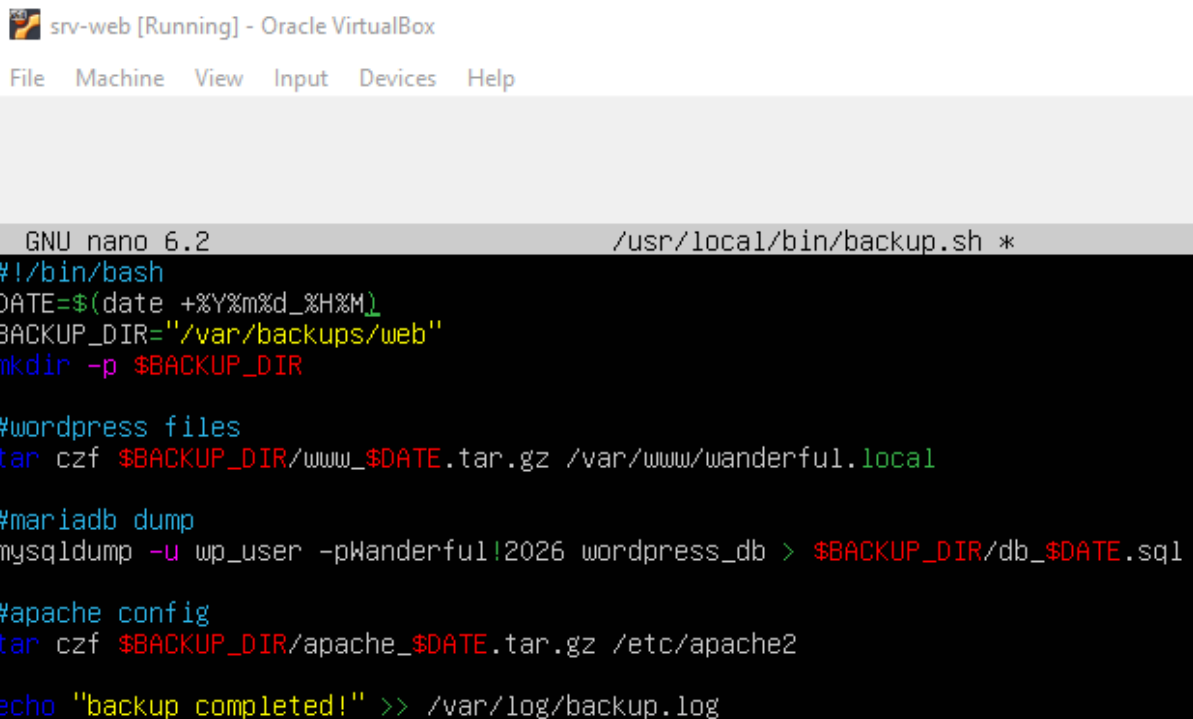
Goal

Create a bash script on srv-web that backs up /var/www, a MariaDB dump, and /etc/apache2.

Decisions

A bash script was written as it is simpler and demonstrates direct knowledge of shell scripting.

/usr/local/bin/backup.sh



The screenshot shows a terminal window titled "srv-web [Running] - Oracle VirtualBox". The menu bar includes "File", "Machine", "View", "Input", "Devices", and "Help". The terminal content shows the GNU nano 6.2 editor editing the file /usr/local/bin/backup.sh. The script content is as follows:

```
#!/bin/bash
DATE=$(date +%Y%m%d_%H%M_)
BACKUP_DIR="/var/backups/web"
mkdir -p $BACKUP_DIR

#wordpress files
tar czf $BACKUP_DIR/www_$DATE.tar.gz /var/www/wanderful.local

#mariadb dump
mysqldump -u wp_user -pWanderful!2026 wordpress_db > $BACKUP_DIR/db_$DATE.sql

#apache config
tar czf $BACKUP_DIR/apache_$DATE.tar.gz /etc/apache2

echo "backup completed!" >> /var/log/backup.log
```

Make executable

```
sudo chmod +x /usr/local/bin/backup.sh
```

5. End-to-End Demonstration

This section traces a path from booting `srv-client` to accessing the Wanderful Travels WordPress website from the Windows browser, demonstrating how each component depends on the previous one.

1. `srv-client` boots and sends a DHCP Discover broadcast. `srv-infra` replies with an IP offer in the range 192.168.56.50–100.
2. `srv-client` receives its IP, gateway (192.168.56.10) and DNS server (192.168.56.10) from `srv-infra`.
3. From `srv-client`: `dig @192.168.56.10 srv-web.wanderful.local` — BIND9 resolves the name to 192.168.56.20.
4. From `srv-client`: `smbclient //192.168.56.10/public -N` — the public Samba share is listed successfully.
5. From the Windows browser, navigate to <https://wanderful.local> — WordPress loads over HTTPS through the VPN.

6. Problems Encountered and Solutions

The following issues were encountered during the implementation of this project.

Problem 1 — DNS resolution not working on srv-client after DHCP

After switching srv-client to DHCP, pinging hostnames like `srv-web.wanderful.local` failed with "Temporary failure in name resolution", even though `dig @192.168.56.10` worked fine. The system resolver was not using srv-infra as its DNS server despite the Netplan configuration. The fix was to replace the stub resolver symlink:

```
sudo rm /etc/resolv.conf
sudo ln -s /run/systemd/resolve/resolv.conf /etc/resolv.conf
sudo systemctl restart systemd-resolved
```

Problem 2 — Apache showing default page instead of WordPress

After enabling the VirtualHost, the browser still showed the Apache default page. The cause was that the configuration file had not been created before running `a2ensite`, so there was nothing to enable. After creating the file at `/etc/apache2/sitesavailable/wanderful.local.conf` with the correct DocumentRoot, disabling the default site and restarting Apache resolved the issue.

Problem 3 — srv-infra accidentally got DHCP addresses

Running `dhclient enp0s8` by mistake on srv-infra caused it to request an address from its own DHCP server, adding two unwanted dynamic IPs alongside the static one. The fix was:

```
sudo dhclient -r enp0s8
sudo netplan apply
sudo ip addr del 192.168.56.105/24 dev enp0s8
```

Problem 4 — Windows could not access Samba shares

File Explorer returned error `0x80004005` when trying to access `\\192.168.56.10\public`, even though ping and SSH to srv-infra worked fine from Windows.

7. Conclusion and Future Improvements

This project implements a functional Linux infrastructure for Wanderful Travels, Lda. across three virtual machines running Ubuntu Server 22.04 LTS. The following services were successfully deployed: static IP addressing, UFW firewall, key-based SSH with ed25519, BIND9 DNS, DHCP, Samba file sharing with public and private shares, and a LAMP stack with WordPress accessible from the Windows browser.

The main lesson learned was the importance of configuring networking before anything else — on multiple occasions, a correctly configured service was unreachable simply due to a missing firewall rule or interface misconfiguration.

Future improvements

- Replace the self-signed HTTPS certificate with a proper CA-signed one
- Make the backup script automated with cron and remote copy via rsync to srv-client.
- Add a health monitoring script writing to /var/log periodically.
- Add fail2ban to block repeated failed SSH attempts on srv-web and srv-infra.